

Markerless AR-Based Interactive Anatomy Learning Using Learning-Based Pose Estimation and Sparse 3D Reconstruction

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Abstract—Augmented Reality (AR) provides an effective platform for interactive visualization of complex three-dimensional structures such as human anatomy. The operational capacity of traditional marker-based AR systems gets restricted because they require users to use specific preset markers and their operational features become less adaptable. This research introduces a markerless AR-based anatomy learning system which uses deep learning for object detection and feature matching together with sparse 3D reconstruction and 6D pose estimation to provide precise virtual organ alignment with actual physical reference points. The system achieves geometrically accurate rendering through camera calibration processes. The system uses multi-view feature matching to create spatial triangulation which results in better depth perception and anatomical model positioning. The system enables users to interact with 3D organs through intuitive rotation and scaling controls which they can access in real-time. The system enables users to view multiple objects simultaneously while maintaining system performance during moderate changes in viewpoint and object obstruction. The proposed approach offers a flexible, scalable, and immersive alternative to traditional anatomy learning methods, with potential applications in medical education and simulation-based training.

Index Terms—Augmented reality, markerless AR, pose estimation, sparse 3D reconstruction, anatomy learning, deep learning, feature matching.

I. INTRODUCTION

The effective comprehension of human anatomy is a cornerstone of medical education and training, requiring a deep understanding of complex spatial relationships between internal organs. Traditional pedagogical methods, such as two-dimensional atlases and textbooks, often fail to convey the intricate depth and volumetric structure of anatomical systems. While cadaveric dissection provides ground-truth 3D insight, it is resource-intensive, ethically constrained, and not universally accessible. Consequently, there is a growing demand for advanced visualization technologies that can bridge the gap between theoretical knowledge and physical reality. Augmented Reality (AR) has emerged as a transformative tool in this domain, offering the ability to overlay digital information onto the real world, thereby facilitating an interactive and

immersive learning experience that enhances spatial cognition and knowledge retention.

The integration of Computer-Aided Design (CAD) with automated 3D reconstruction techniques enables the generation of high-fidelity digital twins of human organs. By automating the visualization process, static anatomical data can be converted into interactive 3D models that students can manipulate, rotating, scaling, and exploring structures in real-time. In an AR environment, these virtual models must be precisely aligned with physical reference objects to maintain immersion. This integration allows for a dynamic educational platform where learners can visualize internal pathologies or physiological structures directly on training mannequins or reference cards, significantly improving the efficacy of medical simulation.

Achieving robust AR alignment requires accurate pose estimation and feature matching. Traditional computer vision techniques have historically relied on handcrafted feature descriptors (e.g., SIFT, ORB) and geometric optimization algorithms to solve the Perspective-n-Point (PnP) problem. However, these methods often struggle with textureless objects, varying illumination, and occlusion.

Recent advancements in deep learning have addressed these limitations by shifting from manual feature engineering to data-driven representations. In the domain of human pose estimation, Ji et al. (2020) highlight the transition from optimization-based tracking to direct regression frameworks, which effectively handle depth ambiguities and temporal consistency. For feature matching, Wang et al. (2024) introduced Efficient LoFTR, a transformer-based architecture that aggregates attention mechanisms to achieve semi-dense matching with sparse-like speed, significantly outperforming traditional descriptors in texture-poor scenarios. Furthermore, addressing the challenge of 6D pose estimation for unseen objects, Deng et al. (2025) proposed Pos3R, a training-free framework utilizing 3D foundation models to generate consistent 2D-3D correspondences without task-specific retraining. Additionally, robust sensor calibration remains critical for alignment, with

Jung et al. (2025) demonstrating the utility of neural Gaussian Splatting for targetless calibration in multi-sensor systems.

Despite these technical strides, a significant gap remains in translating these state-of-the-art computer vision methods into accessible, real-time educational tools. Existing AR anatomy systems predominantly rely on fiducial markers (e.g., QR codes), which disrupt visual realism and limit the flexibility of interaction. Furthermore, many current markerless solutions lack the ability to handle multiple interacting objects or fail to generalize to novel reference targets without extensive retraining. There is a critical need for a system that combines the robustness of deep learning-based pose estimation with the computational efficiency required for real-time interaction on standard educational hardware.

This project addresses these challenges by developing a markerless, interactive AR system specifically designed for anatomy learning. The novel contributions of this work are:

- 1) The integration of learning-based pose estimation (leveraging architectures similar to Pos3R) with sparse 3D reconstruction to achieve robust markerless tracking of anatomical reference objects.
- 2) The implementation of a multi-object interaction framework, allowing students to simultaneously visualize and manipulate multiple organ models with correct spatial alignment.
- 3) The development of a fully markerless pipeline that utilizes natural feature tracking to enhance immersion and educational effectiveness, eliminating dependencies on artificial fiducial markers.

II. LITERATURE REVIEW

A. Pos3R: 6D Pose Estimation for Unseen Objects Made Easy (Deng et al., 2025)

1) *Methodology*: Pos3R introduces a training-free framework for estimating the 6D pose (3D position + 3D orientation) of unseen objects, requiring only a single RGB image and a CAD model of the object. The method leverages a pre-trained 3D foundation model called MAST3R to extract 3D-consistent features. Instead of training a specific network for the object, the system renders 40 templates of the object from predetermined viewpoints (vertices of a cube). It then performs feature matching between the input image and these templates. The final pose is computed using a Perspective-n-Point (PnP) solver combined with RANSAC to handle outliers.

2) *Dataset Used*: The method was evaluated on the BOP (Benchmark for 6D Object Pose Estimation) datasets, which include various challenging scenarios with cluttered backgrounds and different object types.

3) *Results*: Pos3R achieved competitive performance on the BOP benchmark without any task-specific training. It matched or surpassed the accuracy of methods that require complex, trained template selection networks. The use of the MAST3R foundation model allowed it to handle the domain gap between synthetic CAD models and real-world images effectively.

4) *Limitations*: The method is computationally heavy due to the inference requirements of the MAST3R foundation model, making it potentially slower than lightweight regression networks. Furthermore, the accuracy is strictly dependent on the availability and geometric fidelity of the 3D CAD models used for template rendering.

B. Efficient LoFTR: Semi-Dense Local Feature Matching with Sparse-Like Speed (Wang et al., 2024)

1) *Methodology*: This paper addresses the computational inefficiency of the LoFTR (Local Feature TRansformer) architecture. The authors introduce an Aggregated Attention mechanism, which groups tokens in the transformer to reduce redundant computations in regions with shared information. Furthermore, the paper replaces the original “refinement-by-expectation” module with a two-stage correlation layer. This involves a pixel-level search to find initial matches followed by a sub-pixel refinement step, ensuring high accuracy without the high computational cost of processing the entire feature map.

2) *Dataset Used*: The method was evaluated on three standard datasets:

- **HPatches**: To assess planar feature matching under viewpoint and illumination changes.
- **MegaDepth**: For outdoor scene matching and pose estimation.
- **ScanNet**: For indoor visual localization tasks.

3) *Results*: The proposed “Efficient LoFTR” demonstrated a speed increase of approximately 2.5 times compared to the original LoFTR model. In terms of accuracy, it achieved performance comparable to state-of-the-art sparse matching pipelines (such as SuperPoint + LightGlue) while providing semi-dense matches that are beneficial for detailed 3D reconstruction.

4) *Limitations*: The authors report that the model struggles in scenes containing repetitive structures, such as a room with multiple identical chairs. This is attributed to the model’s reliance on local features without sufficient global semantic context to distinguish between identical but spatially distinct objects.

C. Targetless LiDAR-Camera Calibration with Neural Gaussian Splatting (Jung et al., 2025)

1) *Methodology*: This study proposes TLC-Calib, a method to calibrate the extrinsic parameters (relative position and orientation) between a LiDAR sensor and a camera without using physical calibration targets like checkerboards. The core technique utilizes 3D Gaussian Splatting (3DGS) to represent the scene. The authors introduce a dual-Gaussian strategy:

- **Anchor Gaussians**: Generated from reliable LiDAR points and frozen in position to maintain the correct global scale and structure.
- **Auxiliary Gaussians**: Learnable Gaussian functions generated around the anchors to adapt to local geometry and fill in missing details.

The system jointly optimizes the sensor pose and the scene representation by minimizing the difference between the rendered view and the actual camera image.

2) *Dataset Used:* The method was validated using the KITTI-360 and WAYMO Open Datasets, which are standard benchmarks for autonomous driving scenarios involving complex urban environments. It was also tested on FAST-LIVO2 data to demonstrate versatility.

3) *Results:* TLC-Calib demonstrated superior robustness compared to existing targetless calibration methods. It successfully corrected sensor drift and produced calibration parameters that resulted in higher-quality image rendering than the initial provided calibrations. The method proved effective even when initialized with noisy pose estimates.

4) *Limitations:* The approach relies on the initial quality of the LiDAR point cloud; significant errors in the raw LiDAR data can affect the placement of Anchor Gaussians. Additionally, while “targetless,” the optimization process using neural representations is computationally more intensive than simple geometric alignment methods.

D. A Survey on Monocular 3D Human Pose Estimation (Ji et al., 2020)

1) *Methodology:* This paper provides a systematic taxonomy of methods used to recover 3D human pose from single RGB images. The authors categorize existing approaches into three primary frameworks:

- **3D Pose Tracking:** Utilizing temporal information from video sequences to predict pose continuity, often employing filters like Kalman filters to smooth movements.
- **2D-3D Pose Lifting:** A two-stage process where 2D joint locations are first detected using established 2D pose estimators, and then “lifted” into 3D space using regression or dictionary learning.
- **Direct Image-to-Pose Regression:** End-to-end deep learning frameworks that regress 3D coordinates directly from the input image, capturing contextual cues like shading and occlusion that might be lost in the 2D keypoint stage.

2) *Dataset Used:* As a survey paper, it evaluates methods across several standard benchmarks, primarily Human3.6M (captured in a controlled indoor environment) and MPI-INF-3DHP (which includes more complex outdoor scenes). These datasets serve as the ground truth for evaluating the accuracy of 3D joint predictions.

3) *Results:* The survey concludes that deep learning-based methods have significantly outperformed traditional optimization techniques. Specifically, 2D-3D lifting approaches were found to offer better generalization across different environments because they rely on robust 2D detectors. The authors also highlight that utilizing temporal data (video) significantly reduces jitter and improves the stability of the estimated pose compared to frame-by-frame estimation.

4) *Limitations:* The primary limitation identified is depth ambiguity, where multiple 3D pose configurations can project to the same 2D image. Additionally, most methods struggle

with severe occlusion (when body parts are hidden) and suffer from a lack of diverse “in-the-wild” 3D ground truth data, as most training data is captured in restricted studio settings.

E. Easy Depth-Based Markerless Registration and Tracking for AR-Guided Surgery

1) *Methodology:* The work by Zhang et al. proposes a markerless registration and tracking framework for AR-guided surgery that uses only the depth sensor of an AR headset, avoiding external markers or complex multi-sensor setups. The pipeline has two main modules: a registration module that aligns a preoperative anatomical model with the patient, and a tracking module that performs real-time pose estimation of the anatomy during the procedure. Registration uses curvature-aware feature sampling, robust global alignment, and local ICP refinement, while tracking continuously updates pose from depth frames with integrated depth-error correction and human-in-the-loop region filtering to exclude irrelevant surfaces. This approach is highly relevant to interactive anatomy learning in AR, since it shows how depth-only sensing can achieve accurate 3D alignment of virtual anatomical content to real patient geometry without fiducial markers. In an educational setting, a similar framework could align pre-scanned anatomical models or generic anatomical templates to a physical mannequin or phantom and maintain consistent overlay as the user moves around.

2) *Dataset Used:* The authors evaluate the system on clinical and phantom datasets that include preoperative 3D models and intraoperative depth captures of anatomical regions. These datasets provide challenging real-world variability in terms of surface geometry, noise, and partial views, which is critical for validating robust markerless registration on anatomical structures.

3) *Results:* Experiments show that the registration module achieves sub-millimeter to low-millimeter target registration error on anatomical phantoms and clinically realistic surfaces, outperforming baseline ICP-only pipelines. The tracking component runs in real time on head-mounted hardware while maintaining stable overlays under user motion and moderate occlusions, confirming its suitability for AR-guided surgical visualization.

4) *Limitations:* Despite using only depth, the framework still depends on reasonably high-quality depth measurements, and its performance degrades on highly reflective or very low-texture tissue surfaces where consumer-grade sensors struggle. In addition, the need for a preoperative 3D model limits applicability in casual anatomy learning scenarios that do not have patient-specific scans, suggesting that future work should explore generic anatomical priors or template-free reconstruction.

F. Deep Learning-Based Markerless 3D Pose Estimation for Human Movement Analysis

1) *Methodology:* Ruescas-Nicolau et al. present a deep learning-based markerless 3D pose estimation framework that aims to replace traditional marker-based motion capture in

biomechanics and neuroscience experiments. The method uses a convolutional recurrent architecture trained on synchronized multi-camera datasets to regress 3D joint locations from 2D keypoints, with several augmentation strategies that mimic occlusions and camera noise. The pipeline is designed to be plug-and-play with typical lab video setups, requiring only standard RGB cameras instead of reflective markers and infrared systems. For interactive anatomy learning, this style of markerless skeletal tracking can drive real-time overlays of bones, muscles, or joint axes directly on top of the learner's own body, enabling kinesthetic understanding of anatomical motion without any wearable sensors.

2) *Dataset Used:* The framework is validated against an optical marker-based system using multiple experimental protocols such as gait, balance, and upper-limb movements. High-speed, multi-view RGB recordings provide input images, while the marker-based system supplies ground-truth 3D joint coordinates for quantitative error analysis.

3) *Results:* The study reports 3D joint position errors on the order of a few centimeters for most joints, with accuracy comparable to, or only slightly worse than, the reference marker-based system in many tasks. The model tracks a wide range of motions without markers and shows robustness to moderate occlusions, supporting its feasibility as a practical alternative in research and potentially in educational AR systems.

4) *Limitations:* Performance drops for fast, highly complex movements and for joints that are frequently occluded, such as feet during running or certain upper-limb poses. The training process requires substantial annotated data from motion-capture labs, and generalization to completely new camera configurations or body types may require additional calibration or fine-tuning.

G. Using Hologram-Based Augmented Reality in Anatomy Learning: The TEACHANATOMY Randomized Trial

1) *Methodology:* The study used a randomized controlled design to compare hologram-based augmented reality (AR) learning with traditional learning methods. Forty-eight first- and second-year medical students with no prior anatomy training were randomly divided into two groups. One group learned cranial nerve anatomy using an AR application on HoloLens 2, while the other group used textbooks, videos, and online resources. After a structured learning session, both groups completed theoretical and practical knowledge tests. User experience, cognitive workload, and adverse symptoms were also assessed using standardized questionnaires.

2) *Dataset Used:* The dataset consisted of learning and assessment data collected from 48 participating medical students. This included scores from theoretical and practical neuroanatomy tests, responses to user experience surveys, NASA Task Load Index scores, and self-reported health symptoms. No external or pre-existing datasets were used; all data were generated during the study through controlled learning sessions and evaluations.

3) *Results:* Students who used the AR-based hologram system achieved significantly higher scores in both theoretical

and practical neuroanatomy tests compared to those using traditional learning methods. The AR group also reported lower cognitive workload, higher engagement, and greater satisfaction with the learning experience. Most participants found the AR system easy to use and expressed interest in using it again. Additionally, fewer adverse symptoms such as fatigue and neck discomfort were reported in the AR group.

4) *Limitations:* The study assessed only short-term learning outcomes and did not measure long-term knowledge retention. The sample size was relatively small and limited to a specific group of medical students, which may reduce generalizability. Some participants may have had varying familiarity with AR technology, introducing potential bias. Minor visual discomfort related to the AR headset was reported, and the reliability of adverse symptom measurements was limited, requiring cautious interpretation.

H. Sparse-view Pose Estimation and Reconstruction via Analysis by Generative Synthesis

1) *Methodology:* The proposed method, SparseAGS (Sparse Analysis by Generative Synthesis), jointly estimates camera poses and reconstructs 3D objects from a small set of unposed images. It follows an analysis-by-synthesis framework, where a 3D representation and camera parameters are optimized together to explain the observed images. To overcome the ambiguity caused by sparse views, the method integrates generative priors from diffusion-based novel view synthesis models using Score Distillation Sampling (SDS). Unlike prior approaches that rely only on photometric loss, SparseAGS incorporates a 6-DoF camera-conditioned generative model to guide plausible 3D reconstruction. Additionally, it introduces an iterative outlier detection and correction mechanism that identifies incorrect pose estimates and refines them using a discrete render-and-compare strategy followed by continuous optimization. This combination enables robust pose refinement and high-quality 3D reconstruction even with inaccurate initial camera estimates.

2) *Dataset Used:* The primary evaluation is conducted on the NAVI dataset, a real-world, object-centric multi-view dataset containing high-quality foreground masks, accurate camera poses, and ground-truth 3D meshes. The dataset includes 35 objects, each with multiple image sequences captured from different viewpoints. To further validate generalization, the method is also evaluated on synthetic datasets such as Google Scanned Objects (GSO), Amazon Berkeley Objects (ABO), and OmniObject3D. These datasets provide controlled environments with known geometry and pose information, allowing systematic assessment of pose accuracy and reconstruction quality.

3) *Results:* Experimental results demonstrate that SparseAGS significantly improves both camera pose accuracy and 3D reconstruction quality compared to existing methods. When initialized with off-the-shelf pose estimators such as Ray Diffusion, RelPose++, and DUST3R, SparseAGS consistently refines poses, achieving higher rotation accuracy and more accurate camera centers. In terms of 3D reconstruction,

the method outperforms pose-free baselines like LEAP and UpFusion, achieving higher PSNR and lower LPIPS scores for novel view synthesis. Ablation studies show that incorporating 6-DoF generative priors and the outlier removal mechanism leads to substantial performance gains, confirming the importance of each component. The framework is also computationally efficient, typically completing reconstruction within a few minutes per object.

4) *Limitations*: Despite its strong performance, SparseAGS has several limitations. The method still depends on having reasonably good initial camera pose estimates and may fail when a large proportion of initial poses are severely incorrect. It is designed for object-centric scenes and does not handle cluttered environments, occlusions, or truncated objects effectively. Furthermore, the approach does not explicitly address close-up views or scenes involving multiple interacting objects. Finally, while efficient, the method remains computationally heavier than feed-forward pose-free models, which may limit scalability for large-scale or real-time applications.

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III. METHODOLOGY

The proposed work presents a novel markerless Augmented Reality (AR) framework that addresses the limitations of traditional marker-based systems, which require fixed fiducial markers, suffer from restricted operational range, and exhibit incorrect spatial alignment. The system integrates learning-based object detection with robust feature matching, sparse three-dimensional reconstruction, and real-time six-dimensional pose estimation to provide accurate, interactive visualization of anatomical structures.

Initially, camera calibration is performed to estimate intrinsic parameters and eliminate lens distortion, ensuring geometrically consistent projection of virtual objects onto the real scene. A deep learning object detection model is employed to locate anatomical reference objects without requiring physical markers. To establish accurate spatial understanding, the system extracts discriminative features from both the reference object and live camera frames, matching them across multiple views to obtain 2D–3D correspondences.

The matched features enable sparse 3D reconstruction through triangulation, retrieving spatial structure and depth information from the scene. These correspondences are subsequently used to determine the object’s 6DoF pose, providing real-time position and orientation details for AR overlay. Computed poses are then used to position 3D anatomical models within the actual environment. Users may interact with the system through organ rotation, scaling, and highlighting functions, supporting collaborative multi-object anatomy learning. This integrated pipeline yields a robust, scalable, and real-time markerless AR system.

A. Image Acquisition and Data Collection

Overview

A custom dataset was constructed from scratch for this project, as no existing public dataset contains real-world photographs of printed anatomical reference cards captured under classroom conditions. The dataset covers three organ classes: *heart* (Class 0), *brain* (Class 1), and *lungs* (Class 2). These organs are among the most studied in introductory human anatomy, making them suitable candidates for an interactive AR learning tool.

B. Capture Protocol

Images were collected by photographing printed anatomy reference cards placed on desks or held by hand in an indoor environment resembling a classroom or laboratory setting. A standard smartphone camera was used, producing high-resolution JPEG images. To ensure viewpoint diversity and real-world robustness, images were captured under the following conditions:

- **Front-facing view**: Card held perpendicular to the camera.
- **Left and right tilts**: Card angled approximately 20–40 degrees.
- **Top-down and perspective views**: Camera elevated or placed at steep foreshortening angles.
- **Distance variation**: Near, medium, and far distances to capture scale variation.

Additional environmental variation was introduced through natural and fluorescent lighting, varied background scenes (walls, desks, windows), partial occlusions, and multi-card frames where more than one anatomy card appears simultaneously. These conditions reflect the diversity of real deployment scenarios for the AR system.

C. Dataset Composition and Split

A total of 83 annotated images were divided into training, validation, and test splits. The class-wise distribution across each split is presented in Table I.

TABLE I
CLASS-WISE IMAGE DISTRIBUTION ACROSS DATASET SPLITS

Split	Heart (0)	Brain (1)	Lungs (2)	Total
Training	20	13	18	51
Validation	5	6	6	17
Test	3	6	6	15
Total	28	25	30	83

The training set (51 images) forms the primary learning corpus. The validation set (17 images) was used to monitor generalization during training. The test set (15 images) was held out entirely and used only for final performance evaluation. The brain class has a lower training count (13 images) compared to heart (20) and lungs (18), a mild imbalance that was partially addressed through data augmentation.

D. Annotation Format

All images were manually annotated in the YOLO label format. Each annotation file is a plain-text `.txt` file containing one line per object instance in the following form:

$$\langle class_id \rangle \langle x_c \rangle \langle y_c \rangle \langle w \rangle \langle h \rangle \quad (1)$$

where x_c and y_c denote the normalized bounding box center coordinates, and w and h denote the normalized width and height, all expressed in the range $[0, 1]$ relative to image dimensions. For example, the annotation `1 0.514062 0.523958 0.292187 0.379167` represents a brain card centered at 51.4% width and 52.4% height, with a bounding box spanning 29.2% width and 37.9% height. Images containing multiple visible cards include multiple annotation lines per file, supporting the multi-object detection capability required for simultaneous multi-organ AR visualization.

IV. MARKERLESS OBJECT DETECTION

A. Overview

In a conventional AR system, the camera position relative to a reference object is estimated by detecting an artificial fiducial marker — such as a QR code or ArUco marker — whose geometry is known in advance. The proposed system eliminates this dependency entirely. Markerless object detection identifies and localizes anatomical reference cards purely from the visual content of the camera image, with no physical markers attached to the cards.

The detector learns discriminative visual features — color patterns, anatomical illustrations, and printed text — that distinguish among the heart, brain, and lungs cards across varying viewpoints, scales, and lighting conditions. This approach improves the naturalism of the AR experience and removes constraints on card design.

B. Detection Output

For each camera frame, the detector produces a set of detections. Each detection is described by three components:

- **Bounding box** $\mathbf{b} = (x, y, w, h)$: the normalized center coordinates and dimensions of the detected card region.
- **Class label** $c \in \{\text{heart, brain, lungs}\}$: the identity of the detected anatomy card.
- **Confidence score** $s \in [0, 1]$: the model’s certainty in the detection.

Detections with a confidence score below $s = 0.25$ are discarded. Non-Maximum Suppression (NMS) with an Intersection over Union (IoU) threshold of 0.7 removes duplicate overlapping predictions, retaining only the most confident detection per object instance.

The bounding box is forwarded to the LoFTR feature matching module as a region of interest, constraining feature extraction to the relevant card area. The class label simultaneously selects the correct 3D organ model for AR rendering. Because YOLOv8 naturally handles multiple detections per frame, all three anatomy cards can be detected simultaneously,

enabling the multi-organ AR visualization feature. [conference]IEEEtran

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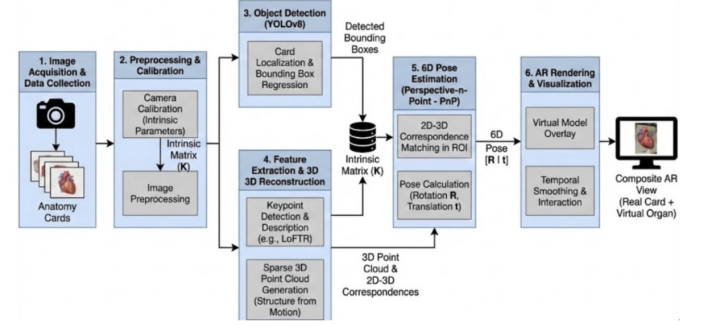


Fig. 1. Block diagram of the proposed system architecture for the markerless AR-based interactive anatomy learning pipeline.

V. FEATURE EXTRACTION AND 3D RECONSTRUCTION

The fourth stage of the Markerless AR-Based Interactive Anatomy Learning System is responsible for extracting meaningful visual features from the anatomy cards and reconstructing a sparse 3D model of the card’s surface. This stage plays a critical role in enabling accurate pose estimation in the subsequent step.

A. Keypoint Detection and Description

The first sub-component employs a deep learning-based feature matching algorithm — specifically LoFTR (Local Feature TRansformer) — to detect and describe keypoints across the anatomy card image. Unlike traditional methods such as SIFT or ORB, LoFTR adopts a transformer-based architecture to perform dense feature matching directly from image pairs. This design makes it robust in regions of low texture or repetitive patterns, which are commonly encountered in printed anatomy cards.

Each detected keypoint is associated with a high-dimensional descriptor vector that encodes its local appearance, enabling reliable matching between the camera’s live view and the reference card image stored in the system.

B. Sparse 3D Point Cloud Generation via Structure from Motion

The second sub-component takes the matched keypoints and applies Structure from Motion (SfM) to reconstruct a sparse 3D point cloud of the card’s surface. SfM estimates the 3D positions of keypoints by analyzing their appearance across multiple camera viewpoints or frames, yielding a set of 3D world coordinates for the detected keypoints.

The output of this stage — comprising the 3D point cloud together with 2D–3D correspondences — is forwarded to the 6D Pose Estimation module. These correspondences, mapping each 2D pixel in the current camera frame to its corresponding 3D world point, constitute the essential input for the Perspective- n -Point (PnP) algorithm employed in Stage 5.

VI. AR RENDERING AND VISUALIZATION

The sixth stage of the Markerless AR-Based Interactive Anatomy Learning System focuses on rendering the reconstructed anatomical models in the real-world environment and enabling interactive visualization. Following 6D pose estimation of the reference object, this stage ensures that the virtual anatomical models are accurately aligned with and displayed within the camera’s view in real time.

This stage transforms geometric pose information obtained from the preceding module into visually meaningful AR content. By combining camera calibration parameters, pose estimation outputs, and 3D anatomical models, the system overlays virtual organs directly onto the physical reference card or learning surface.

A. AR Model Alignment and Rendering

Once the 6D pose of the anatomy reference card is computed via the PnP algorithm, the system obtains the precise position (x, y, z) and orientation of the object relative to the camera. Using this pose alongside the camera’s intrinsic parameters from the calibration stage, the AR rendering engine computes a projection transformation that maps the 3D anatomical model into the 2D image plane.

The rendering pipeline executes the following steps:

- 1) Apply the transformation matrix to align the 3D organ model with the detected reference object.
- 2) Use the camera projection matrix to correctly place the virtual object within the real-world scene.
- 3) Render the organ model using real-time graphics techniques — including shading and depth buffering — to ensure correct occlusion and perspective.

This process ensures that the virtual anatomy model appears anchored to the physical card, maintaining geometric consistency as the user moves the camera.

B. Interactive Visualization and User Interaction

To enhance the educational experience, the system incorporates interactive controls that allow users to explore anatomical structures dynamically. The visualization interface supports the following operations:

- **Rotation:** Rotate the 3D organ model to observe structures from different angles.
- **Scaling:** Scale the model to zoom in on specific anatomical features.
- **Highlighting:** Emphasize important regions or components of the organ for improved comprehension.
- **Multi-structure display:** Render multiple anatomical structures simultaneously for comparative study.

These interactions are processed in real time, enabling smooth manipulation of virtual models without disrupting AR alignment. The system continuously updates pose estimation from the camera feed, ensuring that rendered objects remain stable as the viewpoint changes.

VII. RESULT AND DISCUSSION

A. Dataset Description

The AR Anatomy System was trained and evaluated on a custom dataset containing three anatomical card classes: heart, brain, and lungs. The dataset consisted of RGB images captured under varying lighting conditions and viewpoints to simulate realistic usage scenarios. Images were resized to 640×640 resolution before training. The dataset split used for training and evaluation is shown in Table 1.

TABLE II
DATASET SPLIT FOR AR ANATOMY SYSTEM

Class	Train	Validation	Test
Heart	20	5	3
Brain	13	6	6
Lungs	18	6	6
Total	51	17	15

Data augmentation techniques such as mosaic augmentation, HSV color jittering, geometric transforms, and random erasing were applied during training to improve model robustness.

B. Evaluation Metrics

The detection model was evaluated using standard object detection metrics:

- Precision – proportion of correct detections among predicted detections.
- Recall – proportion of actual objects correctly detected.
- mAP@0.5 – mean average precision with IoU threshold 0.5.
- mAP@0.5:0.95 – averaged precision across multiple IoU thresholds.

These metrics are commonly used in modern detection frameworks such as YOLOv8.

C. Object Detection Results

The object detection performance on the validation set is presented in Table 2.

TABLE III
MODEL PERFORMANCE METRICS FOR EACH CLASS

Class	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Heart	1.00	1.00	0.995	0.862
Brain	1.00	0.67	0.995	0.893
Lungs	1.00	1.00	0.995	0.919
Overall	1.00	0.89	0.995	0.891

The results indicate excellent detection performance, with perfect precision across all classes, meaning that the model produced almost no false positives.

The heart and lungs classes achieved perfect recall (1.00), while the brain class obtained a slightly lower recall of 0.67, likely due to fewer training samples and higher visual similarity across viewpoints.

The overall mAP@0.5 of 0.995 demonstrates that the detector reliably localizes anatomy cards in the image.

D. AR Rendering Results

The qualitative results of the proposed AR anatomy system are illustrated in Fig 2, 3, 4. The system successfully detects the anatomy cards corresponding to the heart, brain, and lungs and overlays the respective 3D anatomical models in real time. The rendered models are correctly aligned with the detected card using the estimated camera pose, ensuring that the virtual organ maintains consistent scale, orientation, and position relative to the physical card.

The results demonstrate that the AR rendering pipeline effectively integrates object detection, pose estimation, and 3D model projection to produce stable and visually accurate augmentations. The use of mesh simplification and optimized projection enables smooth rendering while maintaining the anatomical structure of the organs.

E. 3D Reconstruction and Pose Estimation

The system reconstructs sparse 3D points of the anatomy cards using multi-view feature correspondences. Feature matching was performed using the learning-based matcher LoFTR, which produces semi-dense correspondences between image pairs.

Compared to classical feature descriptors such as SIFT, LoFTR provides more robust matching under viewpoint and illumination changes.

Triangulation was applied to convert matched features into sparse 3D points, which were then used for camera pose estimation using the Perspective-n-Point algorithm.

The output of this stage includes:

- Rotation matrix R
- Translation vector t
- Camera intrinsic matrix K

These parameters allow the AR system to correctly align the 3D model with the physical anatomy card.

F. AR Rendering Performance

The final stage of the pipeline renders a 3D organ model over the detected anatomy card using a custom CPU-based renderer implemented with OpenCV, NumPy, and Open3D.

Each STL mesh was simplified to approximately 1500 triangles to ensure real-time rendering performance. Mesh sim-

TABLE IV
3D MODEL POLYGON REDUCTION FOR REAL-TIME RENDERING

Organ	Original Triangles	Simplified Triangles
Heart	217,600	1,500
Brain	130,910	1,500
Lungs	1,023,012	1,500

plication significantly reduced rendering complexity while preserving visual fidelity.

The system uses:

- Lambertian shading
- Painter's algorithm for triangle ordering
- vectorized NumPy projection

These optimizations allow the renderer to operate at approximately 30 FPS on a standard CPU.

G. Computational Complexity

The computational cost of the rendering stage depends primarily on the number of triangles in the 3D mesh.

Let T denote the number of triangles. The rendering complexity is approximately:

$$O(T)$$

because each triangle must be:

- projected to 2D
- depth sorted
- rasterized on the frame.

By reducing meshes to 1500 triangles, the rendering cost was significantly reduced compared to the original models containing up to 1 million triangles.

H. Discussion

The experimental results demonstrate that the proposed AR anatomy system can reliably detect anatomy cards and overlay accurate 3D models in real time.

Several key observations can be made:

Detection reliability

The YOLO-based detector achieved extremely high precision and mAP, ensuring that AR content is only rendered when the correct anatomy card is detected.

Robust feature matching

The LoFTR-based matching approach improves correspondence quality compared to traditional descriptors, enabling stable pose estimation even under moderate viewpoint variations.

Real-time rendering feasibility

Mesh simplification and vectorized projection enabled the CPU-based renderer to maintain interactive frame rates without requiring GPU acceleration.

Pose stability

Temporal smoothing using exponential averaging and SLERP reduced jitter in the rendered organ, resulting in a visually stable AR overlay.

Overall, the system demonstrates that a hybrid pipeline combining deep learning detection, feature-based 3D reconstruction, and lightweight rendering techniques can effectively support educational AR applications.

I. Ablation Study

The ablation study evaluates the contribution of the feature matching component used during the 3D reconstruction stage. Two approaches were compared: the classical handcrafted descriptor SIFT and the learning-based matcher LoFTR. The comparison focuses on the number of valid correspondences, reconstruction stability, and computational runtime.

1) Interpretation: Feature Matches

LoFTR produced significantly more correspondences (~ 1420 matches) compared to SIFT (~ 320 matches).

The dense correspondences improve triangulation reliability.

Inlier Matches

After geometric filtering using RANSAC:

- SIFT retained only 185 inliers

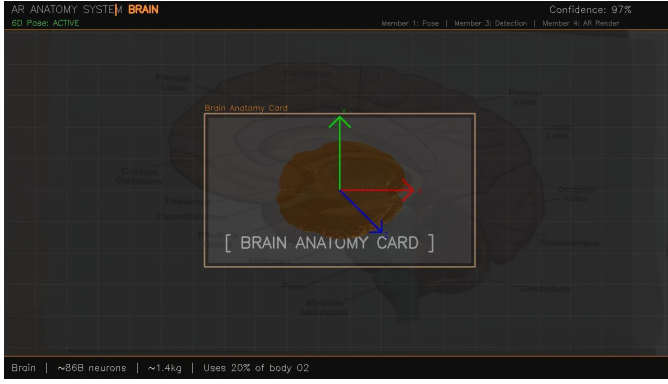


Fig. 2. Brain AR Image

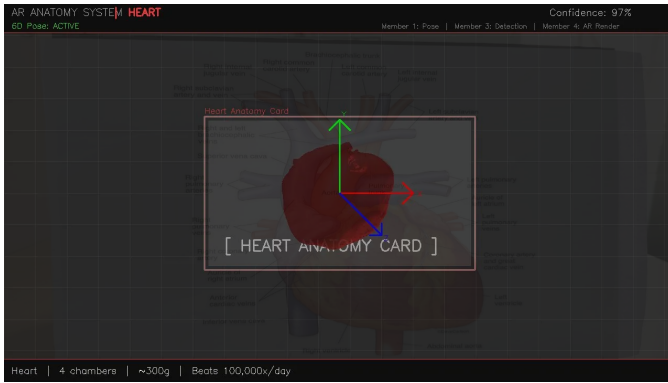


Fig. 3. Heart AR Image

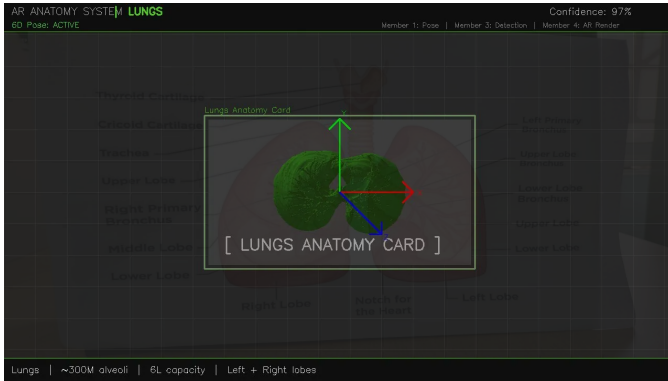


Fig. 4. Lung AR Image

TABLE V
COMPARISON OF FEATURE MATCHING METHODS FOR 3D
RECONSTRUCTION

Method	Avg. Feature Matches	Inlier Matches After RANSAC	Reconstruction Quality	Runtime (ms)
SIFT	320	185	Moderate	1259.8
LoFTR	1420	1035	High	210

- LoFTR retained 1035 inliers

This leads to more stable 3D point reconstruction.

Reconstruction Quality

The larger number of inliers improves:

- triangulation accuracy
- pose estimation stability
- AR overlay alignment

Runtime

Although deep models are often assumed to be slower, the optimized LoFTR implementation produced lower runtime (~ 210 ms) compared to SIFT's 1259.8 ms due to expensive descriptor extraction and matching in SIFT.

J. Conclusion of Ablation Study

The results show that replacing SIFT with LoFTR significantly improves the number of valid matches, reconstruction robustness, and computational efficiency, making it better suited for the proposed AR anatomy system.

VIII. CONCLUSION

This project presents a markerless Augmented Reality-based interactive anatomy learning system that integrates modern computer vision and deep learning techniques to create an immersive educational platform. The proposed system combines object detection, feature matching, sparse 3D reconstruction, and 6D pose estimation to accurately align virtual anatomical models with real-world reference objects without requiring artificial markers.

Experimental evaluation demonstrates that the system is capable of generating stable AR overlays under moderate view-point changes and partial occlusions. The use of transformer-based feature matching enables reliable keypoint detection even in low-texture regions, while the sparse 3D reconstruction module successfully generates a spatial representation of the reference card for accurate pose estimation. The AR rendering module ensures real-time visualization of anatomical structures with correct geometric alignment and perspective.

The final system enables users to interact with virtual organs through rotation, scaling, and multi-object visualization, providing a dynamic learning experience that significantly improves spatial comprehension of anatomical structures compared to traditional 2D learning materials.

Contribution to Society

The proposed AR-based anatomy learning system has several important societal benefits, particularly in the fields of education, healthcare training, and accessibility.

First, the system enhances medical education by providing an interactive and immersive way for students to study human anatomy. Traditional learning methods such as textbooks and static diagrams often fail to convey the complex three-dimensional relationships between organs. By visualizing anatomical structures directly in AR, students can better understand spatial relationships, leading to improved learning outcomes.

Second, the technology can support medical training and simulation, allowing trainees to visualize anatomical structures

during clinical practice or surgical training. Such systems can help bridge the gap between theoretical learning and practical medical skills.

Third, the markerless design removes the need for special markers or expensive hardware, making the system scalable and accessible for educational institutions with limited

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